

17 Type tests, electrical

17.1 General

A sample of completed cable, 10 m to 15 m in length, shall be subjected to the following tests, applied successively:

- a) insulation resistance measurement at ambient temperature (see 17.2);
- b) insulation resistance measurement at maximum conductor temperature in normal operation (see 17.3);
- c) voltage test for 4 h (see 17.4).

Cables of a rated voltage of 1,8/3 (3,6) kV shall also be subjected to an impulse test on a separate sample of completed cable, 10 m to 15 m in length (see 17.5).

The tests shall be limited to not more than three cores.

17.2 Insulation resistance measurement at ambient temperature

17.2.1 Procedure

This test shall be made on the sample length before any other electrical test.

All outer coverings shall be removed and the cores shall be immersed in water at ambient temperature for at least 1 h before the test.

The DC test voltage shall be between 80 V and 500 V and shall be applied for a sufficient time to reach a reasonably steady measurement, but in any case for not less than 1 min and not more than 5 min.

The measurement shall be made between each conductor and the water.

If requested, measurement may be confirmed at a temperature of $(20 \pm 1) ^\circ\text{C}$.

17.2.2 Calculations

The volume resistivity shall be calculated from the measured insulation resistance by the following formula:

$$\rho = \frac{2 \times \pi \times l \times R}{\ln \frac{D}{d}}$$

where

ρ is the volume resistivity, in ohms centimetres;

R is the measured insulation resistance, in ohms;

l is the length of the cable, in centimetres;

D is the outer diameter of the insulation, in millimetres;

d is the inner diameter of the insulation, in millimetres.

The insulation resistance constant, K_i , expressed in megaohms $\times$ kilometres [$\text{M}\Omega \cdot \text{km}$] may also be calculated, using the formula:

$$K_i = \frac{l \times R \times 10^{-11}}{\log \frac{D}{d}} = 10^{-11} \times 0,367 \times \rho$$

NOTE For the cores of shaped conductors, the ratio D/d is the ratio of the perimeter over the insulation to the perimeter over the conductor.

17.2.3 Requirements

The values calculated from the measurements shall be not less than those specified in Table 13.

17.3 Insulation resistance measurement at maximum conductor temperature

17.3.1 Procedure

The cores of the cable sample shall be immersed in water at a temperature within ± 2 K of the maximum conductor temperature in normal operation for at least 1 h before the test.

The DC test voltage shall be 80 V to 500 V and shall be applied for a sufficient time to reach a reasonably steady measurement, but in any case for not less than 1 min and not more than 5 min.

The measurement shall be made between each conductor and the water.

17.3.2 Calculations

The volume resistivity and/or the insulation resistance constant shall be calculated from the insulation resistance by the formulae given in 17.2.2.

17.3.3 Requirements

The values calculated from the measurements shall be not less than those specified in Table 13.

17.4 Voltage test for 4 h

17.4.1 Procedure

The cores of the cable sample shall be immersed in water at ambient temperature for at least 1 h before the test.

A power frequency voltage equal to $4 U_0$ shall then be gradually applied and maintained continuously for 4 h between each conductor and the water.

17.4.2 Requirements

No breakdown of the insulation shall occur.

17.5 Impulse test for cables of rated voltage 1,8/3 (3,6) kV

17.5.1 Procedure

This test shall be performed on the sample at a conductor temperature between 5 K and 10 K above the maximum conductor temperature in normal operation.

The impulse voltage shall be applied in accordance with the procedure given in IEC 60230 and shall have a peak value of 40 kV.

NOTE Electrical tests on repaired installations are subject to installation requirements. The above test is for new installations only.

Table 13 – Electrical type test requirements for insulating compounds

Designation of compounds (see 4.2)	Unit	PVC/A	EPR/ HEPR	XLPE
Maximum conductor temperature in normal operation (see 4.2)	°C	70	90	90
Volume resistivity ρ				
– at 20 °C (see 17.2)	$\Omega \cdot \text{cm}$	10^{13}	–	–
– at maximum conductor temperature in normal operation (see 17.3)	$\Omega \cdot \text{cm}$	10^{10}	10^{12}	10^{12}
Insulation resistance constant K_i				
– at 20 °C (see 17.2)	$\text{M}\Omega \cdot \text{km}$	36,7	–	–
– at maximum conductor temperature in normal operation (see 17.3)	$\text{M}\Omega \cdot \text{km}$	0,037	3,67	3,67